

Review

Problematic Internet Use Among Primary School Children: A Narrative Review of Mental Health and Socioecological Correlates

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Abstract

Background/Objectives: Digital technologies are increasingly embedded in children's everyday lives, yet problematic Internet use (PIU) remains conceptually heterogeneous and has been studied mainly in adolescents. This narrative review aimed to synthesize recent empirical evidence on how Internet use and its problematic manifestations are conceptualized and measured among primary school children and to identify individual and contextual factors associated with the specific Internet-related outcomes assessed from a socioecological perspective. **Methods:** This narrative review was informed by focused, non-exhaustive searches conducted in APA PsycInfo, Web of Science Core Collection, and PubMed for recent quantitative empirical research published between 2021 and 2026 involving primary school populations. Eighteen studies considered directly relevant to the review objectives constituted the central body of literature for the descriptive and interpretive narrative synthesis. **Results:** The studies used heterogeneous terminology, ranging from general or risky Internet use to problematic smartphone and video game use, Internet addiction, and Internet gaming disorder. Of the 18 included studies, 12 used cross-sectional designs, limiting the ability to draw causal inferences from the reported associations. Higher levels or a greater likelihood of problematic use were associated with family conflict, insecure attachment, maladaptive parenting practices, parental technology dependence, socioemotional difficulties, externalizing problems, peer victimization, deviant peer affiliation, and intensive recreational digital use. Positive parent-child relationships, parental supervision, authoritative parenting, parental digital competence, positive peer communication, resilience, traditional games, and a positive school climate were associated with lower levels or a lower likelihood of the corresponding Internet-related outcomes. One of the studies concerning traditional games assessed general Internet use rather than PIU. Within the focused body of literature, only one study provided school-level evidence, and none examined community-level factors. **Conclusions:** PIU may serve as an umbrella term for organizing heterogeneous Internet-related constructs, provided that the conceptual and methodological distinctions among the specific outcomes are preserved. The reported associations span individual and contextual domains, but the predominance of cross-sectional evidence prevents causal interpretation. These findings may inform hypotheses and priorities for developmentally appropriate prevention; however, they should not be



Academic Editor: Ana Adan

Received: 10 August 2026

Revised: 7 September 2026

Accepted: 17 September 2026

Published: 1 October 2026

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regarded as demonstrating the effectiveness of particular interventions. Future research should employ longitudinal designs and more standardized terminology and validated assessment instruments.

Keywords: problematic Internet use; primary school children; socioecological model; risk factors; protective factors; digital media; narrative review

1. Introduction

The expansion of the Internet and digital technologies has facilitated their incorporation into virtually every area of everyday life. Consequently, the daily use of these resources has become common practice among the general population, transforming the ways in which people communicate, learn, engage in leisure activities, and socialize. This phenomenon is particularly relevant during childhood and adolescence, stages in which the Internet, video games, social media, and mobile devices constitute common settings for interaction and entertainment. Although these technologies offer important opportunities for development, their excessive or maladaptive use may have harmful effects on children's well-being. It is therefore essential to conceptually delimit the phenomenon of problematic Internet use.

In this context, problematic Internet use (PIU) is considered an umbrella term encompassing excessive, compulsive, or maladaptive patterns of use of digital technologies associated with online environments [1–3]. This concept includes various problematic behaviors characterized by loss of control and the persistence of the behavior despite its negative consequences, such as Internet gaming disorder, problematic social media use, compulsive online buying, problematic pornography use, cyberchondria, and online gambling [1,3,4]. In children, it is particularly important to distinguish between these subtypes to avoid conceptual confusion, particularly between problematic Internet use, Internet addiction, and problematic social media use [5]. To clarify this conceptual and diagnostic heterogeneity, Table 1 summarizes the principal constructs related to PIU, together with their definitions and current diagnostic status.

Nevertheless, the literature continues to show considerable terminological and conceptual heterogeneity in describing problematic Internet-related behaviors. The absence of unified diagnostic criteria has led to multiple approaches focused on specific behaviors. Internet addiction is not recognized as a distinct formal diagnosis in either the DSM-5-TR or the ICD-11. Internet gaming disorder is included in the DSM-5-TR as a condition for further study, whereas the ICD-11 formally recognizes gaming disorder, encompassing both online and offline gaming [6,7]. This terminological and conceptual heterogeneity makes it difficult to compare studies, estimate prevalence, and develop preventive and therapeutic interventions [8,9]. In this regard, several authors support the usefulness of PIU as an integrative category that encompasses different problematic digital behaviors while acknowledging their specific characteristics. This approach may facilitate greater conceptual consistency and better adaptation to a constantly evolving technological environment [3].

From a transdiagnostic psychiatric perspective, PIU should not be inferred solely from a symptom score or assumed to constitute an independent disorder. Problematic digital behavior may coexist with broader emotional, behavioral, or neurodevelopmental difficulties and may sometimes function as a maladaptive coping response to them. Accordingly, a comprehensive and developmentally appropriate clinical assessment is required to determine its clinical significance, associated functional impairment, psychiatric comorbidity, and differential diagnosis [2,4,10].

Table 1. Definitions and Diagnostic Status of Constructs Related to Problematic Internet Use.

Concept	Definition	Diagnostic Status
Problematic Internet Use (PIU)	An umbrella term describing excessive, compulsive, or maladaptive Internet use associated with negative consequences or functional difficulties across one or more online activities [2,3,5,9].	Not recognized as a distinct diagnosis in the DSM-5-TR or ICD-11 and does not necessarily imply the presence of a clinical disorder [6,7].
Problematic Social Media Use (PSMU)	An activity-specific manifestation of problematic digital use involving impaired control over social media use and associated negative consequences [2–4].	Not recognized as a distinct diagnosis in the DSM-5-TR or ICD-11 [6,7].
Problematic Smartphone Use (PSU)	A maladaptive pattern of smartphone use characterized by impaired control and negative consequences. It may reflect problematic engagement with activities accessed through the device [2–4].	Not recognized as a distinct diagnosis in the DSM-5-TR or ICD-11.
Problematic Video Game Use (PVGU)	A maladaptive pattern of video game use associated with negative consequences that does not necessarily meet the diagnostic criteria for gaming disorder [2–4].	Not recognized as a distinct diagnosis in the DSM-5-TR or ICD-11 [6,7].
Internet Addiction	A research term used to describe persistent addiction-like patterns of generalized Internet use involving loss of control and significant negative consequences [5,9].	Not recognized as a distinct diagnosis in the DSM-5-TR or ICD-11 [6,7].
Internet Gaming Disorder (IGD)	A persistent and recurrent pattern of Internet gaming associated with clinically significant impairment or distress [6].	Included in the DSM-5-TR as a condition for further study rather than as a formally recognized diagnosis [6].
Gaming Disorder (GD)	A pattern of online or offline gaming characterized by impaired control, increasing priority given to gaming, and continuation or escalation despite negative consequences, resulting in significant functional impairment [7].	Recognized as a clinical diagnosis in the ICD-11 [7].

Note. Definitions were synthesized by the authors from the sources cited in the corresponding entries. **Abbreviations:** DSM-5-TR = Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, Text Revision; ICD-11 = International Classification of Diseases, Eleventh Revision.

Although most research on problematic Internet use has focused on adolescence and later developmental stages, childhood—particularly the primary education stage (approximately 6–12 years of age)—represents an especially relevant period for its study. During this stage, digital use habits begin to develop, and some may become established or intensify during later developmental stages. In this regard, restricting the present review exclusively to primary school children avoids grouping together child, adolescent, and adult populations, whose developmental characteristics, psychosocial needs, and patterns of technology use differ considerably. Therefore, identifying the risk and protective factors associated with problematic Internet use at this developmental stage may help inform the future development and evaluation of preventive approaches.

According to Bronfenbrenner’s ecological model [11], children’s behavior can be understood as emerging from interactions among multiple systems of influence. When applied to problematic Internet use, this model suggests that children’s digital behavior may be influenced not only by individual characteristics but also by family dynamics, peer relationships, the school context, and the characteristics of the digital environment [11–13]. Recent empirical studies have reported associations across several of these ecological levels, including emotional and family processes [14], school climate [15], and peer victimization

and peer relationships [16,17]. Figure 1 conceptually situates Internet use during childhood within Bronfenbrenner's interacting ecological levels.

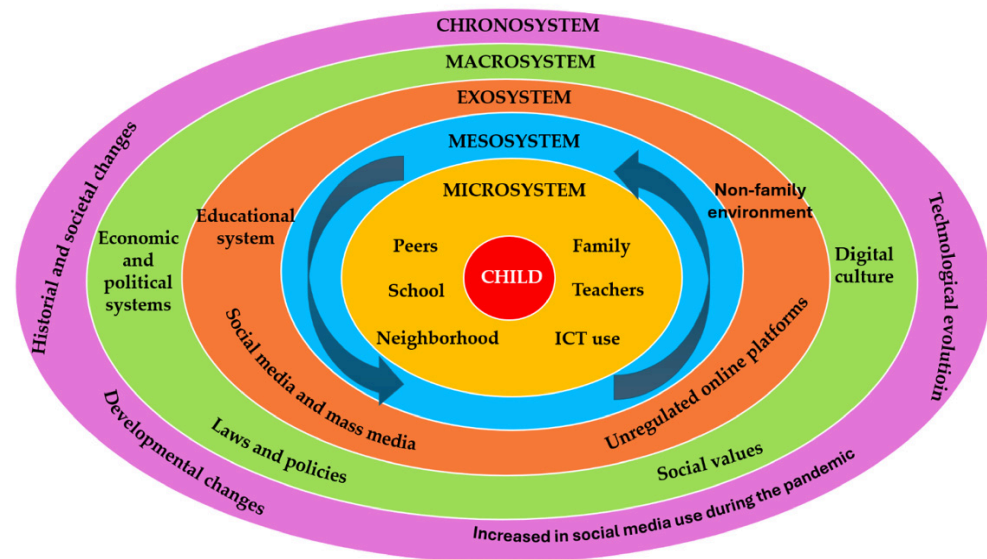


Figure 1. Conceptual Adaptation of Bronfenbrenner's Socioecological Model to Internet Use among Primary School Children. **Note.** Developed by the authors as a conceptual adaptation of Bronfenbrenner's socioecological model. The elements represented illustrate potentially relevant influences across ecological levels and were not necessarily examined in the studies forming the narrative synthesis. Their inclusion and placement should not be interpreted as indicating empirical support within the present review or causal relationships. **Abbreviation:** ICT, information and communication technology.

Despite growing interest in PIU, the evidence concerning primary school children remains conceptually fragmented, with studies employing heterogeneous terminology and frequently examining associated factors within isolated developmental contexts. The present review addresses this fragmentation by jointly examining the conceptualization of PIU and organizing its associated factors across multiple socioecological levels. Accordingly, this narrative review aims to synthesize recent empirical evidence on Internet use and its problematic manifestations among primary school children from a socioecological perspective. Specifically, it seeks to (1) examine how Internet use and its problematic manifestations are conceptualized and measured in the recent empirical studies involving this population and (2) identify the individual, peer, family, school, and digital factors associated with higher or lower levels of the specific Internet-related outcomes assessed. By integrating conceptual analysis with a socioecological organization of the evidence, this review seeks to provide a more comprehensive understanding of PIU during primary education and inform future research and preventive initiatives.

Nevertheless, primary school children should not be regarded as a developmentally homogeneous group. Across this educational stage, substantial differences may exist in cognitive development, emotion-regulation capacities, digital skills, access to devices, and autonomy in Internet use. These differences may influence both the manifestation of PIU and the relevance of its associated risk and protective factors. Therefore, findings referring to primary school children as a whole should be interpreted with appropriate developmental caution.

2. Materials and Methods

This narrative review provides a descriptive and interpretive synthesis of the recent empirical literature concerning the conceptualization and measurement of problematic

Internet use (PIU) and the individual and contextual factors associated with Internet use and its problematic manifestations among primary school children. To inform this synthesis, focused, non-exhaustive searches were conducted in APA PsycInfo, Web of Science Core Collection, and PubMed in June 2026. These databases were purposively selected to provide complementary coverage of the principal areas addressed by the review: APA PsycInfo covers psychological, behavioral, and developmental research; PubMed provides access to the biomedical, psychiatric, and public health literature; and Web of Science offers broader multidisciplinary coverage, including educational and social-science research. The selection of these sources was intended to support a focused narrative synthesis rather than to achieve exhaustive coverage of all potentially relevant publications. Web of Science was selected as the multidisciplinary citation database, and no additional search was conducted in Scopus. The China National Knowledge Infrastructure (CNKI) was not consulted because the review did not include a separate Chinese-language search and focused on studies indexed in the selected international databases rather than on regional database coverage. Consequently, studies indexed exclusively in Scopus, CNKI, or other regional databases may be underrepresented in the focused body of literature considered.

The literature search was organized around three broad concept groups related to: (1) Internet use and its problematic manifestations; (2) individual, interpersonal, and contextual factors; and (3) primary or elementary school children. Search concepts were operationalized primarily using free-text terms drawn from the terminology employed in the recent literature on PIU, child development, and socioecological factors. MeSH descriptors or another single controlled vocabulary were not used as the organizing basis of the search because indexing systems and terminology differ across the three databases and several of the constructs examined lack consistent standardized indexing.

The terms included in the second concept group reflected the predefined socioecological and socioemotional focus of the narrative review and were not intended to constitute an exhaustive list of all factors potentially associated with problematic digital behavior. Because the three concept groups were combined using AND, the search may have preferentially identified studies addressing the selected concepts, while studies focused on other relevant domains may not have been retrieved if they did not also contain one of the selected terms. Accordingly, the present review synthesizes the factors identified within this focused body of literature and should not be interpreted as providing a comprehensive mapping of all possible factors associated with PIU.

Particular attention was given to quantitative empirical research published between 2021 and 2026. This temporal restriction applied specifically to the empirical studies constituting the central body of the narrative synthesis. This temporal focus was chosen because children's Internet use takes place within a rapidly evolving technological context characterized by continuing changes in digital platforms, devices, and patterns of use. It was also consistent with the temporal dimension of Bronfenbrenner's chronosystem and with the aim of examining recent evidence. The temporal restriction did not apply to foundational conceptual, diagnostic, methodological, or theoretical publications used to contextualize the terminology and interpret the findings in the Introduction and Discussion. Earlier publications were therefore consulted when relevant to understanding the development, definition, or theoretical interpretation of PIU and related constructs, but they were not included among the 18 recent empirical studies forming the central body of the synthesis. The central body of literature for the narrative synthesis comprised 18 studies considered directly relevant to the review objectives. Accordingly, the analysis of conceptualization and measurement focused on their treatment within this recent empirical corpus and was not intended to provide a comprehensive historical review of the conceptual development of PIU.

The terminology used to describe Internet use and its problematic manifestations was examined comparatively, whereas the associated factors were interpreted according to the direction and context of the reported associations and their socioecological level. The terms risk factor and protective factor were retained only when referring to the terminology employed in the original publications. In the narrative synthesis, variables examined in cross-sectional studies were described as factors associated with higher or lower levels of the specific Internet-related outcome assessed rather than as evidence of risk or protection. Differences in study design, sample characteristics, informant or data source, measurement approaches, and temporal structure were considered when interpreting the evidence. Study-specific effect estimates were descriptively charted in Table 2 where available. However, given the substantial heterogeneity in outcome definitions, measurement instruments, statistical models, effect-size metrics, reference groups, and covariate adjustment, the reported effect estimates were not considered directly comparable and were therefore not summarized as numerical ranges. Findings derived from cross-sectional studies were interpreted as associations rather than as evidence of causal relationships.

Given the narrative design of the review, no formal risk-of-bias instrument or overall methodological-quality rating was applied. Instead, methodological characteristics relevant to the interpretation of the evidence—including study design and temporal structure, sample characteristics, informant or data source, measurement approach, and statistical reporting—were considered descriptively during the interpretive synthesis. This descriptive consideration should not be interpreted as constituting a formal risk-of-bias assessment.

3. Results

The results of the 18 studies included in this narrative review are presented in four sections. First, the main study characteristics are summarized, including their geographical distribution, research design, and the Internet-related outcomes and constructs assessed. Second, the terminology used to classify and conceptualize Internet use and its problematic manifestations in childhood is examined. Third, variables associated with higher levels or a greater likelihood of the specific Internet-related outcomes assessed are synthesized across individual, family, peer, school, and digital domains. Finally, variables associated with lower levels or a lower likelihood of these outcomes are presented across the relevant socioecological domains. Given the heterogeneity in terminology, outcome measures, and study designs, the findings are reported through narrative synthesis and are interpreted primarily as associations rather than causal relationships. The factors synthesized in the third and fourth sections are those represented in the focused body of literature identified through the concept-guided search and should not be interpreted as a comprehensive catalogue of all factors potentially associated with Internet use or its problematic manifestations.

The included studies did not assess a single homogeneous outcome. They examined general Internet use, risky Internet use, generalized problematic or compulsive Internet use, Internet addiction, problematic smartphone use, problematic video game use, and Internet gaming disorder. These constructs capture different dimensions of children's digital behaviour, including use intensity or duration, exposure to online risks, device- or activity-specific problematic use, impaired control, addiction-like symptoms, and functional impairment, and should not be regarded as interchangeable outcome measures. Throughout the Results section, the terminology used in each original study is retained when reporting its findings. PIU is used collectively in this review only as an umbrella term and should not be interpreted as indicating equivalence among these outcomes. Accordingly, associations identified for one specific construct were not assumed to generalize to the other forms of Internet-related behaviour.

For organizational purposes, the expressions “risk factors” and “protective factors” have been retained as broad headings reflecting the terminology used in the original publications. Within these sections, however, findings derived from cross-sectional studies are described according to the direction of the reported association—that is, as factors associated with higher or lower levels of the specific Internet-related outcome assessed. In this context, these terms do not imply temporal precedence, causality, or a demonstrated risk-enhancing or protective effect.

Table 2 summarizes the main characteristics of the 18 studies constituting the central body of the narrative synthesis, including authorship and year of publication, country, sample size, participants’ age and school grade, study design and temporal structure, informant or data source, the specific Internet-related outcome assessed, and the corresponding assessment instrument. Where reported in the original publication, the table also identifies the instrument’s age-related validation or intended population, scoring approach, cutoff value, and whether the measure assessed frequency or duration of use, exposure to online risks, impaired control, addiction-related symptoms, or functional impairment. The principal findings and study-specific effect estimates are also presented where available.

Table 2. Characteristics, Outcome Definitions, and Main Findings of the Studies Included in the Narrative Synthesis.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Miltuze et al. [18] Latvia.	N = 261 children aged 8–11 years; M = 8.55, SD = 2.02; 138 girls, 123 boys; 2nd–3rd grade at T1. N = 236 child–parent dyads at T2.	Longitudinal; 2 measurement waves; 1-year follow-up.	Child and parent reports.	Compulsive Internet Use (CIU); Compulsive Internet Use Scale (CIUS).	Child–parent relationship; parenting practices; Internet-related parenting practices; parental restrictions/rules. Family/individual.	CIU at T1 predicted CIU at T2 ($\beta = 0.38$, $p < 0.001$). At T1, parental rules were negatively associated with CIU ($\beta = -0.26$, $p < 0.001$), whereas forbidding ($\beta = 0.24$, $p < 0.001$) and controlling ($\beta = 0.17$, $p < 0.01$) were positively associated. At T2, child–parent relationship was negatively associated with CIU ($\beta = -0.17$, $p < 0.01$), while forbidding ($\beta = 0.16$, $p < 0.01$) and inconsistent parenting ($\beta = 0.13$, $p < 0.05$) were positive predictors. Model $R^2 = 0.28$.
Przepiorka et al. [19] Poland.	N = 427 primary-school students; 10–14 years; M = 13.06, SD = 1.09; 218 girls, 209 boys.	Cross-sectional.	Child self-report.	Problematic smartphone use (PSU); Smartphone Addiction Scale.	Social anxiety; social skills; somatization; academic performance. Individual.	Social skills negatively predicted PSU in girls ($\beta = -0.31$, $p < 0.001$) and boys ($\beta = -0.17$, $p = 0.020$). PSU negatively predicted GPA in girls ($\beta = -0.19$, $p = 0.006$) and positively predicted somatization in boys ($\beta = 0.30$, $p < 0.001$). Partial mediation was reported for social skills: GPA through PSU in girls and social skills; somatization through PSU in boys.
Kacar & Ayaz-Alkaya [20] Türkiye.	N = 42 fifth and sixth grade students; intervention N = 20; control N = 22.	Quasi-experimental pretest–posttest with control group; 8-week traditional-games intervention.	Child self-report.	Internet addiction and Internet-use duration; Family–Child Internet Addiction Scale.	Traditional children’s games; social skills; perceived stress. Individual/digital.	No baseline between-group differences were found in Internet addiction, Internet use, social skills, or stress ($p > 0.05$). At post-test, the intervention group showed lower daily Internet use ≥ 3 h/day (10.0% vs. 45.5%; $\chi^2 = 6.453$, $p = 0.011$) and lower daily-use frequency (20.0% vs. 59.1%; $\chi^2 = 6.645$, $p = 0.010$) than controls. Parent-reported social skills were higher in the intervention group (281.75 vs. 253.31; $U = 127.50$, $p = 0.020$), whereas perceived stress did not differ (6.55 vs. 7.59; $U = 181.00$, $p = 0.324$).

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Pons-Salvador et al. [21] Spain.	N = 1827 parents of children aged 6–9 years.	Cross-sectional.	Parent report.	Children’s Internet use and parental mediation; assessment focused on frequency/use and parental mediation rather than a problematic-use construct.	Parental digital competence; parental supervision and knowledge of children’s online activities. Family/digital.	Higher parental Internet use was strongly associated with parental digital competence [$F(1,1660) = 378.34, p < 0.001,$ $\eta^2 = 0.19$]. Having a social-network profile was associated with higher digital competence [$W(1758.02) = 107.01,$ $p < 0.001, d = 0.60$]. Parents who always supervised their children’s Internet use showed higher digital competence than those who never did ($d = 0.28, p < 0.001$). Parents who knew their children’s online activities also had higher competence ($d = 0.49, p = 0.001$).
Yun et al. [22] South Korea.	N = 342 children; 9–10 years; participants from the Panel Study on Korean Children.	Two-wave longitudinal secondary analysis; 2017–2018; random-effects panel logistic model.	Child, primary caregiver, and teacher reports, depending on variable.	Problematic smartphone use; K-Scale (Internet Addiction Diagnostic Scale), modified by replacing “Internet” with “smartphone, media.”	Externalizing problems; internalizing problems; happiness; self-esteem; peer communication; parental supervision; authoritative, authoritarian and permissive parenting; school adaptation. Individual/family/ peers/school.	Externalizing problems were associated with greater PSU ($OR = 3.14, 95\% CI$ $[1.25, 7.87], p = 0.015$), as was permissive parenting ($OR = 6.68, 95\% CI [1.89,$ $23.56], p = 0.003$). Peer communication ($OR = 0.42, 95\% CI [0.20, 0.89], p = 0.023$), parental supervision ($OR = 0.41, 95\% CI$ $[0.19, 0.87], p = 0.020$), and authoritative parenting ($OR = 0.13, 95\% CI [0.04, 0.39],$ $p = 0.001$) were protective.
Bautista-Alcaine et al. [23] Spain.	N = 319 fifth- and sixth-grade students; M age = 10.58; 52.4% boys.	Cross-sectional; gender-differentiated linear regression analyses.	Child self-report.	Problematic Internet Use (PIU); standardized questionnaire measure of PIU.	Weekday and weekend Internet-use time; family supervision; problematic video-game use. Digital/family.	For girls, problematic video game use ($\beta = 0.342$), weekday Internet use ($\beta = 0.128$) and family supervision ($\beta = -0.023$) were predictors of PIU; $R^2 = 0.504$. For boys, problematic video game use ($\beta = 0.501$) and weekday Internet use ($\beta = 0.105$) were predictors; $R^2 = 0.625$. PIU correlated with problematic video game use ($r = 0.59$ girls; 0.70 boys) and weekday Internet use ($r = 0.47$ girls; 0.59 boys).

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Zhou et al. [24] China.	N = 65,868 fourth-grade students; M age = 9.56, SD = 0.62; 54.0% boys.	Cross-sectional; data collected September–October 2015; moderated mediation PROCESS Model 15).	Child self-report.	Internet addiction (IA); Internet Addiction Diagnostic Questionnaire, Chinese version.	Peer victimization; subjective well-being; parent–child relationship; socioeconomic status. Peers/school/ individual/family.	Peer victimization positively predicted IA ($\beta = 0.265$, 95% CI [0.256, 0.274]). Subjective well-being negatively predicted Internet addiction ($\beta = -0.119$, 95% CI [−0.127, −0.111]) and mediated 12.03% of the total association between peer victimization and Internet addiction. Parent–child relationship moderated both the direct peer victimization → Internet addiction path ($\beta = -0.017$, 95% CI [−0.025, −0.008]) and the subjective well-being → Internet addiction path ($\beta = -0.034$, 95% CI [−0.042, −0.026]).
Long et al. [25] China.	N = 433 parents of elementary-school students; 80 fathers, 353 mothers; children from grades 1–6.	Cross-sectional; structural equation modelling with mediation and moderation (PROCESS Models 4 and 59); Conducted during COVID-19 lockdown.	Parent report.	Internet addiction; parent-reported child Internet-addiction measure.	Parental smartphone dependence; parent–child conflict; parental role. Family/digital.	Parental smartphone dependence was positively associated with children's Internet addiction ($\beta = 0.52$, $t = 12.77$, $p < 0.001$, 95% CI [0.44, 0.60]) and parent–child conflict ($\beta = 0.37$, $t = 8.78$, $p < 0.001$, 95% CI [0.29, 0.46]). Parent–child conflict was positively associated with Internet addiction ($\beta = 0.29$, $t = 6.82$, $p < 0.001$, 95% CI [0.21, 0.38]). Indirect effect = 0.11; 17.5% of the total effect was mediated. Parental role moderated both mediation paths.
Olmeda-Muelas et al. [26] Spain.	N = 641 children; M = 10.15, SD = 0.89.	Cross-sectional; mediation analysis.	Child self-report.	Problematic Internet Use (PIU); EUPI-a, 11-item Problematic Internet Use Scale.	Parental attachment security; maladaptive cognitive emotion-regulation strategies: self-blame, rumination, catastrophizing and other-blame. Family/individual.	Problematic users represented 34.3% ($n = 220$). PIU correlated negatively with mother attachment ($r = -0.46$) and father attachment ($r = -0.42$), and positively with rumination ($r = 0.44$), catastrophizing ($r = 0.40$), other-blame ($r = 0.23$), and self-blame ($r = 0.16$), all $p < 0.01$. Mother attachment showed an indirect effect through maladaptive regulation strategies ($B = -0.13$, 95% CI [−0.16, −0.09]); father attachment showed $B = -0.11$, 95% CI [−0.14, −0.07]. Rumination and catastrophizing were the most consistent mediators.

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Ayaz-Alkaya & Akca [27] Türkiye.	N = 314 fifth- and sixth-grade students; M age = 11.58, SD = 0.49; 56.1% girls.	Cross-sectional.	Child self-report.	Internet use; researcher-developed questionnaire assessing duration, frequency, device and purpose of Internet use (not a problematic-use outcome).	Traditional children's games; social skills; stress; gender and academic level. Individual/digital.	Children who did not play traditional games were more likely to use the Internet (OR = 2.16, 95% CI [1.17, 3.99], <i>p</i> = 0.014). Social skills were associated with daily Internet-use time (β = 0.164, <i>p</i> = 0.003), traditional gaming (β = 0.155, <i>p</i> = 0.008), and weekly gaming frequency (β = 0.123, <i>p</i> = 0.035). Stress was associated with gender (β = 0.120, <i>p</i> = 0.033) and academic level (β = 0.111, <i>p</i> = 0.048).
Derin et al. [28] Türkiye.	N = 142 children; 6–12 years; M = 8.5, SD = 1.7; 65% girls; 71 children with ADHD and 71 controls.	Cross-sectional; ADHD/control comparison; path analysis.	ADHD diagnosis: Parent and child clinical assessment/interviews; RIU: parent-report questionnaires.	Risky Internet Use (RIU); Parent–Child Internet Addiction Test (PCIAT-20).	Formal ADHD diagnosis (K-SADS-PL-DSM-5); social skills; demographic covariates. Individual.	ADHD was associated with poorer social skills (β = −1.68, <i>p</i> < 0.001), and poorer social skills were associated with greater RIU (β = −0.57, <i>p</i> = 0.004). The direct ADHD–RIU association was nonsignificant (β = −0.52, <i>p</i> = 0.169), whereas the indirect association through social skills was significant (β = 0.96, <i>p</i> = 0.004). The cross-sectional design does not establish causal mediation.
Liu et al. [29] China.	N = 556 fourth- to sixth-grade students; M age = 10.30, SD = 0.93; 53.4% boys; grade 4 = 31.3%, grade 5 = 34.7%, grade 6 = 34.0%.	Cross-sectional; sequential mediation; 5000 bootstrap samples	Child self-report.	Internet addiction; 20 item Internet Addiction Test (IAT).	Emotional abuse; emotional dysregulation; regulatory emotional self-efficacy. Family/individual.	Emotional abuse was positively associated with Internet addiction (r = 0.37, <i>p</i> < 0.001). Emotional abuse predicted emotional dysregulation (β = 0.48, <i>p</i> < 0.001), regulatory emotional self-efficacy (β = −0.17, <i>p</i> < 0.01), and Internet addiction (β = 0.23, <i>p</i> < 0.001). Indirect effects through dysregulation = 0.14, 95% CI [0.072, 0.210]; self-efficacy = 0.03, 95% CI [0.007, 0.067]; sequential pathway = 0.02, 95% CI [0.008, 0.053].

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Theopilus et al. [30] Indonesia.	N = 1627 parents of children aged 7–11 years.	Cross-sectional parent survey; partial least squares-structural equation modelling (PLS-SEM)	Parent report.	Internet addiction risk; Parent–Child Internet Addiction Test (PCIAT).	Parental mediation: active, co-use, restrictive-general, restrictive-specific and supervision; mobile/non-mobile device use; learning/productivity and recreational activities. Family/digital.	All five parental mediation strategies were negative predictors of IA: active $\beta = -0.119$; co-use $\beta = -0.060$; restrictive-general $\beta = -0.216$; restrictive-specific $\beta = -0.150$; supervision $\beta = -0.181$. Mobile-device use ($\beta = 0.284$) and recreational activities ($\beta = 0.157$) were positive predictors, whereas learning/productivity activities were negatively associated ($\beta = -0.068$). Model $R^2 = 0.529$.
Wei & Xu [31] China.	N = 419 children; M age at Wave 1 = 10.70, SD = 0.58; 54.4% boys.	Two-wave longitudinal study; 1-year interval; moderated mediation.	Child self-report.	Internet gaming addiction; 9 item Internet-gaming-disorder questionnaire.	School climate; deviant peer affiliation; sensation seeking. School/peers/individual.	School climate at T1 was negatively correlated with Internet gaming addiction at T2 ($r = -0.29$, $p < 0.001$). School climate predicted lower deviant peer affiliation ($b = -0.11$, SE = 0.04, $p < 0.01$), which predicted higher gaming addiction ($b = 0.29$, SE = 0.06, $p < 0.001$). Indirect effect = -0.033 , 95% CI [-0.063 , -0.009]. Sensation seeking moderated the deviant peer affiliation $\rightarrow$ gaming addiction association ($b = 0.10$, SE = 0.04, $p < 0.05$).
Zhou et al. [32] China.	N = 65,868 fourth-grade students. Further sample characteristics were NR in the accessible article information.	Cross-sectional; moderated mediation model.	Child self-report. exact data source NR.	Internet addiction (IA); instrument NR from accessible article information.	Peer relationship; peer victimization; resilience. Peers/individual.	Peer victimization partially mediated the negative association between peer relationship and IA. Resilience showed a protective moderating effect across the mediation pathways, with stronger protection among children with high resilience. Exact coefficients, confidence intervals and instrument details are NR because the full article was not accessible.

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Derin et al. [33] Türkiye.	N = 120 children; 6–12 years; M = 8.73, SD = 1.80; 58% boys; 60 first-time ADHD diagnoses and 60 controls.	Cross-sectional case–control design; dual first- and second-stage moderated mediation (PROCESS Model 58).	Parent- and child-report measures; formal ADHD diagnosis based on clinical assessment; maternal childhood trauma was mother-reported.	Risky Internet Use (RIU); Parent–Child Internet Addiction Test–20-Item Version (PCIAT-20).	Formal ADHD diagnosis; parent–child conflict; maternal childhood trauma; covariates including child sex/age, parental education and maternal depression/anxiety. Individual/family.	ADHD was associated with parent–child conflict ($b = 0.751$, 95% CI [0.42, 1.08]), and conflict was associated with RIU ($b = 0.417$, 95% CI [0.22, 0.62]). Conditional indirect associations through conflict were 0.165 at low maternal trauma, 0.313 at mean trauma, and 0.488 at high trauma. The study therefore found a stronger indirect ADHD–RIU association through parent–child conflict at higher levels of maternal childhood trauma.
Liu et al. [34] China.	N = 661 fourth- to sixth-grade students; M age = 11.23, SD = 0.77; 49.5% boys; grade 4 = 4.4%, grade 5 = 50.7%, grade 6 = 44.9%.	Cross-sectional; moderated sequential mediation (PROCESS Model 88).	Child self-report.	Internet addiction; 20-item Internet Addiction Test (IAT).	Authoritative, authoritarian and permissive parenting; self-control; rejection sensitivity; achievement motivation. Family/individual.	Authoritative parenting had a negative total association with Internet addiction ($\beta = -0.17$, $p < 0.05$), whereas authoritarian parenting had a positive association ($\beta = 0.17$, $p < 0.05$); permissive parenting was non-significant ($\beta = 0.07$, $p > 0.05$). Self-control negatively predicted Internet addiction ($\beta = -0.26$, $p < 0.001$), rejection sensitivity positively predicted it ($\beta = 0.22$, $p < 0.001$) and achievement motivation negatively predicted it ($\beta = -0.50$, $p < 0.01$). Achievement motivation buffered risk and enhanced protective pathways.

Table 2. Cont.

Study [Author et al. (Year) [Ref.]] Country	Sample Characteristics [N; Age Range; M (SD); Grades; Sex/Gender]	Study Design and Temporal Structure [Design; Waves; Follow-Up]	Informant or Data Source [Child/Parent/Teacher/ Clinical/Secondary Data]	Internet-Related Outcome and Assessment Instrument [Exact Outcome— Instrument]	Associated Factors, Assessment, and Socioecological Domain [Factor—Assessment— Domain]	Main Findings and Reported Effect Estimates [Direction; Effect Estimate; CI/ <i>p</i> ; Adjusted/Unadjusted]
Tannous et al. [35] Lebanon.	N = 892 parents with children aged 6–10 years.	Cross-sectional; mediation and serial mediation analyses.	Parent report.	Child problematic media use (PMU); Problematic Media Use Measure (PMUM), 9-item parent-report short form.	Parental smartphone addiction; parent–child conflict; parental psychological distress; parent–child closeness. Family/digital.	Parental smartphone addiction directly predicted child PMU ($\beta = 0.30$, 95% bootstrap CI [0.26, 0.34], $p < 0.001$). Total indirect effect $\beta = 0.12$, CI [0.08, 0.16]; conflict-only indirect effect $\beta = 0.09$, CI [0.06, 0.12]; serial pathway through psychological distress and conflict $\beta = 0.02$, CI [0.01, 0.03]. Total effect $\beta = 0.39$, CI [0.35, 0.43].

Note. The terminology used for each Internet-related outcome or construct corresponds to that employed in the original study and does not imply equivalence among the different outcomes. Information concerning instrument scoring, cutoff values, validation or intended use for the relevant age group, and measurement focus is reported only when explicitly provided in the original publication. Measures assessing primarily the frequency or duration of Internet use were not considered equivalent to measures assessing impaired control, negative consequences, or functional impairment. Effect estimates are reproduced as reported by the original authors and should not be compared directly because of differences in outcome definitions, measurement instruments, statistical models, reference groups, and covariate adjustment. The methodological characteristics reported in the table are intended to support critical interpretation and do not constitute formal risk-of-bias assessments or ratings of methodological quality. Formal ADHD diagnoses and questionnaire-based ADHD-related symptoms are reported separately. Findings derived from cross-sectional studies represent associations and should not be interpreted as evidence of causal relationships. **Abbreviations:** ADHD, attention-deficit/hyperactivity disorder; CI, confidence interval; IGD, Internet gaming disorder; NA, not applicable; NR, not reported; OR, odds ratio; PIU, problematic Internet use.

3.1. Study Characteristics

The studies included in this review were conducted across a diverse range of countries: six studies were carried out in China [24,25,29,31,32,34], four in Turkey [20,27,28,33], three in Spain [21,23,26], and one study each in Latvia [18], Lebanon [35], South Korea [22], Indonesia [30], and Poland [19].

Regarding study design, 12 studies employed cross-sectional designs [19,21,23–27,29,30,32,34,35]. Two studies employed case–control designs [28,33]. Three studies employed longitudinal designs [18,22,31], one of which was a cohort study [22]. Finally, one study employed a quasi-experimental pretest–posttest design with a control group [20].

Participants' ages and school grades varied substantially across the studies, and the evidence included children from both the earlier and later years of primary education. However, several studies combined participants from multiple ages or grades and reported only aggregated findings, while others did not provide sufficiently detailed age- or grade-specific results. Consequently, the available evidence could not be reliably characterized as deriving predominantly from either younger primary school children or preadolescents, nor could the associations be synthesized according to narrower developmental stages. Table 2 reports the age range, mean age, and school grade provided in each original publication, where available.

Regarding the topics addressed, the studies used heterogeneous terminology to conceptualize children's digital behavior. Seven studies assessed Internet addiction [20,24,25,29,30,32,34]. In addition, four constructs were each examined in two studies: Internet use [21,27], risky Internet use [28,33], problematic smartphone use [19,22], and problematic Internet use [23,26]. Finally, four topics were examined in one study each: children's compulsive Internet use [18], child problematic media use [35], problematic video game use [23], and Internet gaming disorder [31]. Because Bautista-Alcaine et al. [23] examined both problematic Internet use and problematic video game use, the number of thematic classifications exceeds the total number of included studies.

The informant or data source used in each study is reported in the Study Characteristics table. Most studies relied on questionnaires completed directly by children, whereas others used parent-reported information concerning children's Internet use, behavior, or family context. A smaller number incorporated clinical assessments, diagnostic information, or secondary panel data. This variation represented an additional source of methodological heterogeneity because these data sources reflect different perspectives and should not be regarded as directly equivalent. Children's self-reports, particularly at younger ages, may be influenced by differences in comprehension, recall capacity, and awareness of their own behavior. Conversely, parental reports may be affected by parents' perceptions, their knowledge of their children's online activities, and their opportunities to observe such behavior. The informant or data source was therefore considered when interpreting the associations reported across studies.

Measurement approaches also varied substantially across the included studies. Some studies assessed the frequency or duration of Internet use or exposure to online risks, whereas others employed screening instruments addressing impaired control, addiction-related symptoms, negative consequences, or functional impairment. Cutoff values and evidence concerning the age-related validation or suitability of the instruments were not consistently reported. Consequently, measures based primarily on use frequency or duration were not considered equivalent to instruments assessing impaired control or functional impairment, and elevated screening scores were not interpreted as establishing a clinical diagnosis. These measurement characteristics are reported in Table 2 where the corresponding information was available in the original publication.

3.2. Classification and Conceptualization of PIU in Childhood

The studies included in this review used heterogeneous terminology to assess Internet use and its various problematic manifestations in childhood, reflecting the lack of conceptual consensus in this field.

One group of studies examined general or risk-related patterns of Internet use employing the terms Internet use [21,27] and risky Internet use [28,33]. These studies focused primarily on the frequency or intensity of digital technology use and on exposure to online-risk behaviors.

A separate group of studies assessed problematic forms of use, employing terms such as children's compulsive Internet use [18], child problematic media use [35], problematic Internet use [23,26], problematic smartphone use [19,22], and problematic video game use [23]. These constructs refer to excessive or maladaptive patterns of use without necessarily implying the presence of a clinical disorder.

Another group of studies examined constructs framed in terms of addictive behaviors, including Internet addiction [20,24,25,29,30,32,34] and Internet gaming disorder [31]. These terms were retained as reported in the original studies and should not be interpreted as equivalent diagnostic categories.

3.3. Risk Factors Associated with Problematic Internet Use

3.3.1. Family and Digital Factors

Eight studies identified family-related variables associated with higher levels of or a greater likelihood of problematic Internet use [18,22,25,26,29,33–35]. These included emotional abuse [29], parental smartphone dependence or addiction [25,35], inconsistent parenting practices characterized by a lack of clear rules [18], authoritarian parenting [34], permissive parenting [22], indirect pathways involving permissive parenting [34], insecure attachment to parents [26], and parent–child conflict [25,33,35].

At the digital level, two studies reported associations between patterns of Internet using and problematic Internet use [23,30]. Recreational online activities, including social networking, gaming, and watching videos or films, were associated with higher levels of Internet addiction [30]. Similarly, problematic video game use and more time spent online on school days were associated with higher levels of problematic Internet use [23].

3.3.2. Individual and Peer Factors

At the individual level, eight studies identified associations between problematic Internet use and various personal and socioemotional variables [19,22,24,26,28,29,31,33]. These included social anxiety symptoms [19], poorer social skills [19,28], lower subjective well-being [24], clinically diagnosed ADHD [28,33], dimensionally assessed externalizing problems [22], emotional dysregulation [29], maladaptive cognitive emotion regulation strategies, including rumination, catastrophizing, and other-blame [26], and higher sensation seeking, which strengthened the association between deviant peer affiliation and Internet gaming addiction [31].

At the peer level, three studies identified associations between problematic Internet use and interpersonal variables, including poorer peer relationships and peer victimization [24,32] and deviant peer affiliation [31]. In the latter study, the association between deviant peer affiliation and Internet gaming addiction was significant among children with higher levels of sensation seeking.

3.4. Protective Factors Associated with Problematic Internet Use

3.4.1. Family and School Factors

Eight studies identified family-related variables associated with lower levels or a lower likelihood of problematic Internet use [18,21–24,26,30,34]. These included positive parent–child relationships [18,24], restrictive parental mediation [30], parental supervision [22,23,30], secure parental attachment [26], authoritative parenting [34], and greater parental digital competence [21].

At the school level, a positive school climate was indirectly associated with lower levels of Internet gaming addiction over time through reduced deviant peer affiliation [31].

3.4.2. Individual, Behavioral, and Peer Factors

Four studies identified individual, behavioral, and peer-related factors associated with lower levels of problematic Internet use [20,22,24,32]. These included greater subjective well-being [24], participation in traditional children’s games [20], positive communication with peers [22], and higher resilience, which buffered the association between peer relationship difficulties and Internet addiction [32].

Additionally, Ayaz-Alkaya and Akca [27] found that more frequent participation in traditional children’s games was associated with shorter Internet-use sessions and less daily Internet use. However, this study assessed general Internet use rather than problematic Internet use.

To provide an integrated visual synthesis of the findings, Figure 2 presents a two-panel fishbone summary distinguishing factors associated with higher levels or a greater likelihood of the specific Internet-related outcomes assessed from those associated with lower levels or a lower likelihood of these outcomes across the socioecological domains represented in the included studies.

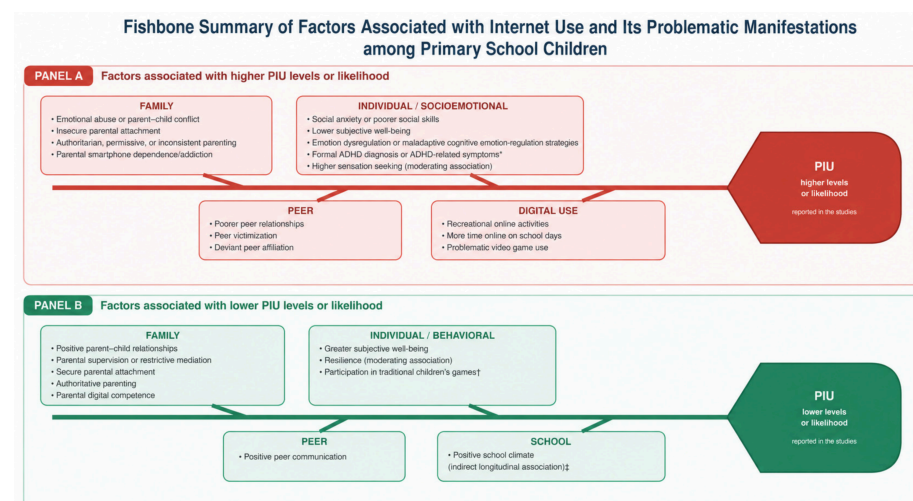


Figure 2. Fishbone Summary of Factors Associated with Internet Use and Its Problematic Manifestations among Primary School Children. **Note.** The diagram summarizes the direction of associations with the specific Internet-related outcome assessed in each included study. The placement of a factor does not indicate that it was associated with every construct included under the umbrella term PIU, and the fishbone structure and arrows do not imply causal relationships. * Formal ADHD diagnoses and questionnaire-based ADHD-related symptoms were reported in different studies and were not treated as methodologically equivalent. † One study addressing participation in traditional children’s games assessed general Internet use rather than PIU. ‡ In one longitudinal model, school climate was indirectly associated with Internet gaming addiction via deviant peer affiliation; this indirect association should not be interpreted as evidence of causal mediation. No community-level factors were examined in the included studies. **Abbreviations:** ADHD, attention-deficit/hyperactivity disorder; PIU, problematic Internet use.

4. Discussion

The present narrative review aimed to examine how problematic Internet use (PIU) and related constructs were conceptualized and measured in the recent empirical studies considered and to synthesize the factors associated with higher or lower levels of the specific Internet-related outcomes assessed among primary school children. Overall, the identified constructs were heuristically organized into broad conceptual categories rather than interpreted as sequential stages along a continuum of severity. The reported associations spanned individual, family, peer, school, and digital domains. Consistent with the ecological model [36], these domains may interact across children's developmental contexts; however, the evidence synthesized in this review does not establish their interdependence or causal relationships. Accordingly, the discussion first addresses the classification and conceptualization of PIU in childhood, followed by the factors associated with higher or lower levels of the relevant Internet-related outcomes across the different socioecological domains. The clinical and practical implications, limitations, and directions for future research are then considered.

4.1. Classification, Conceptualization, and Transdiagnostic Psychiatric Perspective on PIU in Childhood

This review revealed considerable terminological heterogeneity in descriptions of children's digital behavior, reflecting a lack of consensus regarding the conceptualization of problematic Internet use (PIU), a limitation widely acknowledged in the literature [3,5,8]. The included studies used a wide range of terms to describe different forms of engagement with digital environments, ranging from general patterns of Internet use [21,27] to problematic and addiction-related manifestations [18–20,22–26,28–35]. Although many of these constructs overlap substantially, they can be differentiated according to the specific digital behavior to which they refer, and the level of severity attributed to them.

In this regard, children's digital behaviour is described using related but conceptually distinct constructs whose boundaries may overlap. Although the previous literature has sometimes organised different forms of Internet use along a continuum of severity [37,38], this framework should not be interpreted as implying a necessary developmental or clinical progression. Internet-use intensity, exposure to online risks, impaired control, negative consequences, and functional impairment represent distinct dimensions that may occur independently or in different combinations. High engagement or exposure to online risks therefore does not, by itself, constitute an intermediate stage leading to PIU or addiction-related patterns. Figure 3 presents a heuristic conceptual organization of the Internet-use constructs discussed in this review.

One of the categories represented is adaptive Internet use, understood as functional use that is compatible with adequate daily functioning and consistent with children's developmental needs. From this perspective, time spent online does not, in itself, constitute an indicator of problematic use but should instead be interpreted in relation to its context [37,39].

A conceptually distinct category comprises excessive Internet use and risky online behaviors. These patterns may increase children's vulnerability to potentially harmful situations in digital environments, such as sharing personal information or interacting with strangers [40]. However, the evidence suggests that there is no clearly defined boundary between these risk-related behaviors and more problematic patterns of use, further supporting a dimensional understanding of the phenomenon [38].

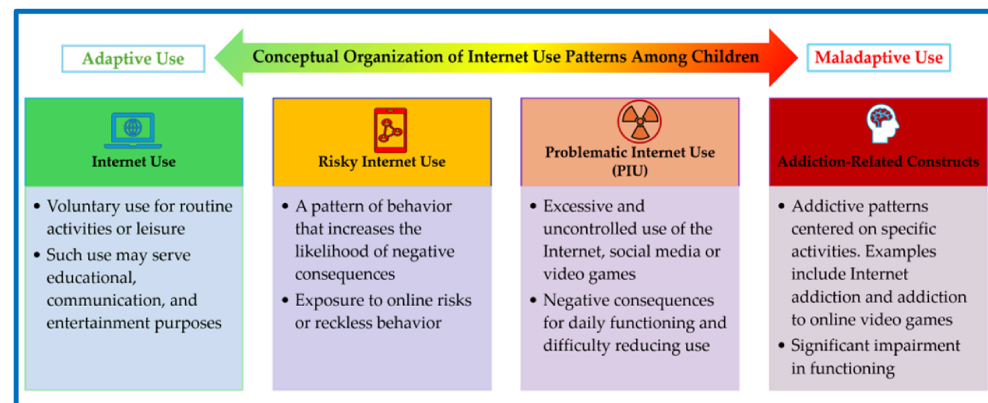


Figure 3. Conceptual Organization of Internet-Use Constructs among Children. **Note.** The left-to-right arrangement provides a heuristic organization of the constructs and should not be interpreted as representing sequential stages or an inevitable developmental or clinical progression. The categories may occur independently or overlap and are not intended to be mutually exclusive or diagnostically equivalent. Exposure to online risks, intensity of engagement, impaired control, negative consequences, and functional impairment represent distinct dimensions. Consequently, high engagement or exposure to online risks should not, by themselves, be interpreted as indicating PIU or an addiction-related pattern. **Abbreviation:** PIU, problematic Internet use.

Problematic Internet use represents another group of constructs within this conceptual organization. PIU is understood as an umbrella term encompassing excessive or maladaptive patterns of use associated with negative consequences for personal, social, family, or academic functioning, without necessarily implying the presence of an addictive disorder [5]. This category includes constructs such as problematic Internet use, problematic smartphone use, problematic social media use, and problematic video game use, which may share underlying mechanisms but differ in the specific digital activity through which they are manifested [3].

A further category comprises constructs framed in terms of addictive behavior, such as Internet addiction and Internet gaming disorder. These constructs refer to patterns characterized by a significant loss of control, clinically relevant functional impairment, and symptoms resembling those of behavioral addictions [41,42]. However, these terms should not be interpreted as equivalent formal diagnostic categories. Internet addiction is not recognized as a distinct diagnosis in either the DSM-5-TR or the ICD-11. Internet gaming disorder is included in the DSM-5-TR as a condition for further study, whereas the ICD-11 formally recognizes gaming disorder [6,7].

Taken together, these findings reveal considerable terminological and conceptual diversity in the literature on children's digital behavior. This heterogeneity hinders comparisons across studies and supports the usefulness of problematic Internet use (PIU) as an umbrella term capable of integrating different problematic manifestations of digital behavior while preserving their specific characteristics [2,8].

From a transdiagnostic psychiatric perspective, elevated scores on PIU screening instruments should not be interpreted as evidence of an independent psychiatric disorder. Rather, problematic digital behavior may represent a heterogeneous clinical manifestation arising through processes shared across diagnostic categories, including emotion dysregulation, impulsivity, negative affect, social withdrawal, difficulties in self-regulation, and maladaptive coping [1,2,4,41]. It may coexist with, exacerbate, or function as a response to emotional, behavioral, or neurodevelopmental difficulties. Pediatric clinical evidence has identified associations between PIU and depressive disorders, attention-deficit/hyperactivity disorder, autism spectrum disorder, sleep disturbances, and functional impairment, although the direction and mechanisms of these relationships remain unclear [43–46]. Recent

adolescent research has also identified bridge symptoms linking Internet addiction and depression, particularly the use of the Internet to escape from problems and difficulties with concentration [45].

Consequently, PIU questionnaires should be understood as screening instruments rather than diagnostic tools. A clinically meaningful assessment should examine the specific online activity involved, impaired control, persistence despite negative consequences, functional impairment, motives for use, and the temporal relationship between problematic use and other psychiatric symptoms. It should also incorporate developmental and family history, information from multiple informants, and assessment of mood, anxiety, attention-deficit/hyperactivity, neurodevelopmental, trauma-related, and, when clinically indicated, psychotic-spectrum symptoms or emerging maladaptive personality patterns. Consideration of psychotic and personality-related symptoms should be developmentally sensitive in primary school children. Without such an assessment, it is not possible to determine whether PIU represents a primary behavioral problem, a co-occurring condition, a coping response to broader psychopathology, or a bidirectional process. Interventions focused exclusively on reducing Internet use could therefore overlook the child's underlying clinical needs.

Accordingly, the classification and clinical interpretation of PIU should not be based exclusively on symptom severity. Several authors have proposed incorporating motives for use and the ecological context in which Internet use occurs, recognizing that children's digital behavior emerges from dynamic interactions among multiple individual, family, school, and social systems throughout development [37,47,48]. From this perspective, the heuristic conceptual organization presented in Figure 3 may be considered alongside an ecological analysis of the personal and contextual factors associated with higher or lower levels of the specific Internet-related outcomes assessed.

4.2. Risk Factors Associated with Problematic Internet Use

4.2.1. Family and Digital Factors

The primary school years represent an important developmental period during which the family context constitutes an important setting for children's socioemotional development and the acquisition of healthy habits. However, the evidence synthesized in this review indicates that family-related variables may also be associated with a greater likelihood of problematic Internet use. Specifically, the included studies associated PIU with emotional abuse, parental smartphone dependence or addiction, inconsistent parenting practices characterized by a lack of clear rules, authoritarian parenting, direct or indirect pathways involving permissive parenting, insecure parental attachment, and parent-child conflict.

These findings are consistent with previous evidence indicating that dysfunctional family relationships and lower-quality emotional bonds are associated with higher levels of problematic Internet, social media, and video game use [49–52]. Such family contexts may be characterized by insecure attachment, conflictual or abusive interactions, and inconsistent or authoritarian parenting practices, which may hinder family functioning and the establishment of consistent guidance regarding technology use [53]. In these circumstances, children may turn to the Internet to compensate for unmet emotional needs or manage emotional distress arising from the family environment, consistent with compensatory models of problematic Internet use [41,54].

The role of digital technology is not limited to children's behavior but also extends to parents' own technology use. Parental computer and Internet use may shape children's digital behavior through monitoring, mediation, and behavioral modeling [55]. Because parents are important behavioral models during childhood, frequent parental smartphone

or Internet use, combined with limited mediation or permissive digital parenting, may contribute to the normalization of high levels of digital engagement and potentially to the development of maladaptive use patterns among children [56–58].

Regarding digital factors, this review found that certain patterns of use were associated with higher levels of PIU, particularly recreational online activities and more time spent online on school days [23,30]. These findings are consistent with the previous literature suggesting that risk is associated not only with the amount of time spent online but also with the type, purpose, and context of the digital activity [59]. From this perspective, concern may arise not from digital activities themselves, but when their use displaces other experiences that are important for child development, such as play, physical activity, sleep, or face-to-face social interactions [37,41].

In particular, problematic video game use was associated with higher levels of PIU in one of the included cross-sectional studies [23]. This association may be interpreted in relation to the immediate rewards, continuous progression, and recurrent reinforcement characteristic of many online games, which may facilitate the development and maintenance of compulsive patterns of use among vulnerable individuals [41,42,60]. This finding is compatible with locating gaming-related problems within the broader spectrum of problematic Internet-related behaviors. However, problematic video game use and Internet gaming disorder should not be treated as interchangeable constructs.

4.2.2. Individual and Peer Factors

Two included case-control studies examined children with clinically established ADHD diagnoses [28,33]. Both studies reported indirect statistical associations between ADHD status and risky Internet use through poorer social skills [28] or greater parent-child conflict [33], respectively. These indirect paths were estimated from cross-sectional data and therefore do not establish temporal ordering or causal mediation. By contrast, Yun et al. [22] did not assess an ADHD diagnosis but examined externalizing problems, including hyperactivity/inattention and conduct-related difficulties, as dimensional questionnaire scores in a panel study. These findings should be considered separately because questionnaire-derived behavioral or symptom scores are not equivalent to a diagnosis established through formal clinical assessment. Yun et al. [22] found that higher externalizing-problem scores were associated with a greater likelihood of problematic smartphone use.

Taken together, these findings suggest that socioemotional difficulties may represent markers of vulnerability to PIU. First, poorer social skills and social anxiety may be associated with greater reliance on online interaction. One possible explanation is that children with fewer social competencies may perceive online interactions as safer and more controllable than face-to-face relationships. This preference may be associated with a greater likelihood of developing problematic patterns of Internet use, particularly when digital environments are used to meet social needs that remain unmet offline [61–63].

Second, emotion regulation was identified as another potentially relevant associated process. The included studies associated emotional dysregulation and maladaptive cognitive emotion regulation strategies with higher levels of PIU [26,29]. More broadly, recent evidence suggests that children and adolescents experiencing greater difficulty managing negative emotions or higher levels of psychological distress may be more likely to use the Internet as a coping or emotion-regulation strategy. However, the direction of this relationship remains unclear, and the available longitudinal evidence does not support a uniformly unidirectional pathway from emotional difficulties to PIU [43].

From a psychobiological perspective, psychosocial stress may also involve changes in stress-responsive biological systems relevant to emotional functioning and self-regulation. Recent pediatric evidence has linked cumulative social adversity to higher concentrations of

multiple inflammatory biomarkers and a nonlinear pattern of cortisol levels [64]. Similarly, lower family affluence has been associated with greater cortisol production and shorter leukocyte telomere length among European children aged 5–12 years [65]. These findings are consistent with the possibility that contextual adversity may become biologically embedded during childhood.

Evidence directly examining biomarkers in relation to problematic digital behaviors among adolescents remains limited and mixed. Higher salivary cortisol was associated with greater problematic social media use among adolescents with depression [66], whereas another clinical study found no significant differences in diurnal cortisol or salivary alpha-amylase profiles between male adolescents with Internet gaming disorder and healthy controls [67]. Taken together, these preliminary findings suggest that stress-related neuroendocrine and inflammatory processes warrant further investigation as potential mechanisms linking contextual adversity, emotion-regulation difficulties, and susceptibility to problematic digital engagement. However, the available evidence is predominantly cross-sectional and does not establish whether biomarker differences precede problematic use, result from it, or reflect other factors, such as depression, sleep disturbance, or socioeconomic adversity. Moreover, none of the studies included in the present review directly assessed biomarkers in primary school children. Biological markers should therefore be considered a promising direction for future research rather than established mechanisms of PIU.

Peer relationships also appear to play an important role. In the included studies, poorer peer relationships, peer victimization, and deviant peer affiliation were associated with higher levels of problematic Internet-related behaviors [24,31,32]. Peer victimization was also associated with lower subjective well-being. In the cross-sectional model reported by the study, lower subjective well-being statistically accounted for part of the association between peer victimization and Internet addiction [24]. This indirect association should not be interpreted as demonstrating temporal or causal mediation.

Clinically established ADHD diagnoses [28,33] and dimensionally assessed externalizing problems, including hyperactivity/inattention [22], were examined in different studies and should not be treated as methodologically equivalent. Both were associated, directly or indirectly, with higher levels or a greater likelihood of the specific Internet-related outcome assessed. Attention difficulties, impulsivity, and self-regulation difficulties may make highly stimulating digital activities that provide immediate gratification particularly appealing. Some children with ADHD-related symptoms may also use digital environments to compensate for difficulties in their offline peer relationships. Nevertheless, the available evidence indicates that associations between ADHD symptoms and digital media use may be reciprocal rather than exclusively unidirectional [44,46].

Finally, sensation seeking should be interpreted as a moderating variable rather than as an independent risk factor in the included evidence. Wei and Xu [31] found that the association between deviant peer affiliation and Internet gaming addiction was significant among children with higher levels of sensation seeking. This moderating pattern is consistent with theoretical models emphasizing interactions between personal predispositions and contextual influences [41], as well as with previous research linking sensation seeking to online gaming addiction [68].

Taken together, the associations summarized within the organizational framework of this review should be interpreted from a reciprocal and multifactorial perspective. ADHD-related symptoms, emotional dysregulation, social anxiety, poorer peer relationships, and lower well-being may increase children's vulnerability to problematic digital engagement. Conversely, problematic digital use may contribute to or exacerbate some of these difficulties, while shared underlying processes—such as self-regulation difficulties, broader psychopathology, or adverse family and social contexts—may contribute to both. The pre-

dominance of cross-sectional evidence prevents these possibilities from being distinguished. Moreover, longitudinal evidence indicates that associations involving emotional difficulties and ADHD-related symptoms may be reciprocal, change over time, or differ according to the outcome examined [43,44]. Therefore, the placement of a variable within a higher- or lower-association category indicates the direction of the reported statistical association and should not be interpreted as establishing its temporal position or causal role.

4.3. Protective Factors Associated with Problematic Internet Use

4.3.1. Family and School Factors

The evidence synthesized in this review indicates that children's access to and use of digital technologies during the primary school years may be associated with characteristics of the family context. Research conducted across childhood and adolescence suggests that parents influence Internet use through rules, behavioral modeling, and mediation practices [69,70]. In this regard, the review identified authoritative parenting, parental supervision and mediation, positive parent-child relationships, secure attachment to parents, and greater parental digital competence as family-related variables associated with lower levels of or a lower likelihood of PIU [18,21–24,26,30,34].

These findings are broadly consistent with previous evidence suggesting that parental mediation operates within a broader family context characterized by secure attachment, open communication, emotional support, a positive family climate, and clear and consistent rules [49,51,63,71]. However, the evidence does not indicate that active parental mediation is uniformly effective, as its associations with children's digital outcomes may vary according to the mediation strategy and outcome examined. From this perspective, supervision should not be understood solely as a strategy for controlling or restricting use, but also as an educational process through which parents guide children in developing digital skills and online self-regulation [72,73]. Such guidance may support the development of more autonomous and responsible use of digital environments over time. Accordingly, learning to use the Internet adaptively and internalizing appropriate boundaries may be associated with a lower likelihood of developing maladaptive patterns of use [37]. Parental guidance may also be relevant to broader online-safety outcomes, including involvement in cyberbullying [74,75]. Taken together, these findings highlight the family as an important potential protective context during this developmental stage.

At the school level, one longitudinal study reported that a positive school climate was associated with lower levels of Internet gaming addiction one year later, with lower deviant peer affiliation included as an indirect statistical pathway in the model [31]. This finding is broadly consistent with previous research suggesting that a safe school environment, characterized by positive student-teacher relationships and a strong sense of school belonging, is associated with children's emotional well-being and psychosocial adjustment [76–79]. A positive school climate may also facilitate healthy peer relationships and provide opportunities for face-to-face interaction [76,79]. School climate and connectedness have also been associated with lower levels of problematic smartphone and social media use, although some of this evidence comes from adolescent samples [80]. Therefore, these findings support the relevance of the school context, while indicating that further research specifically involving primary school children is needed.

4.3.2. Individual, Behavioral, and Peer Factors

The evidence synthesized in this review identified several personal, behavioral, and peer-related variables associated with lower levels of or a lower likelihood of problematic Internet use. Specifically, positive peer communication [22], participation in traditional children's games [20], and resilience [32] were associated with lower levels or a lower

likelihood of the corresponding Internet-related outcomes. Ayaz-Alkaya and Akca [27] also found that participation in traditional games was associated with lower levels of general Internet use; however, this study did not assess problematic Internet use and therefore provides only indirect evidence.

Social skills may also constitute a relevant socioemotional resource. However, Przepiorka et al. [19] examined the relationships among social skills, problematic smartphone use, somatic symptoms, and academic performance using a cross-sectional mediation model; consequently, its findings should not be interpreted as demonstrating that stronger social skills independently protect against PIU. The previous literature has nevertheless identified associations between social competence and problematic or addictive Internet-related behaviors [63]. According to Caplan's model [61], psychosocial difficulties may contribute to a preference for online rather than face-to-face social interaction. From this perspective, stronger offline social skills could facilitate satisfactory face-to-face relationships and reduce reliance on online environments to meet social needs. Nevertheless, this explanatory mechanism was not directly tested in the study by Przepiorka et al. [19].

Positive peer communication was also associated with a lower likelihood of problematic smartphone use [22]. Communication characterized by mutual respect, listening, and the discussion of concerns and problems may provide children with emotional and social support, potentially reducing the need to use the Internet as a compensatory means of social interaction. This interpretation is consistent with previous evidence showing that positive peer relationships are associated with lower levels of PIU and more favorable socioemotional outcomes. However, this previous evidence primarily concerns adolescents, and the overall association between peer relationships and PIU appears to be relatively small [81].

Participation in traditional children's games was also examined as a potentially relevant behavioral factor. In the quasi-experimental study by Kacar and Ayaz-Alkaya [20], an eight-week traditional-games intervention was followed by reductions in daily and weekly Internet use and improvements in social skills compared with the control group. In addition, Ayaz-Alkaya and Akca [27] found that children who participated in traditional games reported shorter Internet-use sessions and less daily Internet use. However, the latter study assessed general Internet use rather than PIU and used a cross-sectional design. These activities may offer healthy leisure alternatives that promote physical activity, motor development, and face-to-face social interaction [82]. Nevertheless, reductions in Internet or screen time should not be interpreted as equivalent to reductions in PIU.

Finally, resilience was identified as a potentially relevant moderating variable. Zhou et al. [32] found that the association between peer relationship difficulties and Internet addiction was weaker among children reporting higher levels of resilience. One possible interpretation is that resilience may be associated with more adaptive coping and less reliance on the Internet for emotional regulation, although this pathway was not directly established. This interpretation is consistent with previous evidence showing a negative association between resilience and PIU [83]. However, this relationship should not be interpreted as causal, as most of the studies synthesized in that meta-analysis used cross-sectional designs.

4.4. Limitations and Future Research

This narrative review provides a focused descriptive and interpretive synthesis of recent evidence concerning the conceptualization of PIU and the factors associated with it among primary school children.

Nevertheless, several limitations should be considered when interpreting the findings. First, as is generally the case with narrative reviews, there is an inherent risk of

bias in the selection and interpretation of the included studies. No formal risk-of-bias instrument was applied, and the findings were therefore not weighted according to an overall methodological-quality rating. Instead, methodological characteristics relevant to their interpretation—including study design and temporal structure, sample characteristics, informant or data source, measurement approach, and statistical reporting—were considered descriptively. Although this approach increases transparency, it does not substitute for a formal risk-of-bias assessment, and individual findings should not be regarded as equally robust.

A further limitation concerns the focused and concept-guided search strategy. The second concept group included selected socioemotional and contextual variables aligned with the predefined scope of the narrative review but did not encompass every domain potentially associated with problematic digital behavior. Moreover, combining the three concept groups using AND may have excluded relevant studies that did not contain one of the selected terms. Consequently, domains such as ADHD or impulsivity, depression and anxiety, sleep, parent–child conflict, attachment, parenting style, bullying, family psychopathology, and socioeconomic conditions may be underrepresented. Scopus, CNKI, and other regional databases were not consulted, and studies indexed exclusively in these sources may therefore be underrepresented or absent. The factors synthesized in this review should consequently be understood as those identified within the focused body of literature consulted rather than as a comprehensive catalogue of all factors associated with PIU. Similarly, the absence of a particular factor from the present synthesis should not be interpreted as evidence that it has not been examined in the broader literature.

The 2021–2026 temporal restriction represents a related limitation. Although earlier foundational conceptual, diagnostic, methodological, and theoretical publications were consulted to contextualize the recent evidence in the Introduction and Discussion, the synthesis of study-specific terminology and measurement approaches is based on the recent empirical corpus. It should therefore not be interpreted as a comprehensive historical review of the conceptual development of PIU.

A further limitation concerns the methodological characteristics of the studies constituting the central body of the narrative synthesis. Most of the studies employed observational designs, predominantly cross-sectional, limiting the ability to establish causal relationships and understand the developmental course of problematic Internet use during childhood. Only three longitudinal studies—including one cohort study—and one quasi-experimental study were identified. Future research should therefore make greater use of longitudinal designs to examine the stability and developmental trajectories of PIU, as well as the prospective relationships between factors associated with higher or lower levels of the relevant Internet-related outcomes across development.

The descriptive nature of the synthesis also limits conclusions regarding the relative magnitude and clinical significance of the identified associations. Although study-specific effect estimates are reported in Table 2 where available, the included studies used heterogeneous effect-size measures based on different constructs, instruments, statistical models, reference groups, and adjustment variables. Their coefficients could therefore not be meaningfully summarized as a common numerical range. Consequently, the presence and direction of an association should not be interpreted as indicating that the corresponding factor necessarily has a strong or clinically meaningful effect.

Some of the included studies used cross-sectional models to decompose total associations into direct and indirect statistical effects. In cross-sectional designs, however, this decomposition represents a statistical partition of covariance under an assumed model and does not demonstrate that the proposed mediator causally transmits an effect. Such analyses cannot establish temporal precedence, exclude plausible alternative models, or

provide evidence of causality. Accordingly, the indirect effects reported by the included studies are interpreted in this review only as statistical associations consistent with the proposed pathways, rather than as evidence of mediation in a causal sense. Longitudinal or experimental designs involving temporally ordered measurements are required to evaluate whether these variables operate as causal mediating mechanisms.

An additional limitation is that most of the included studies assessed PIU and associated psychological difficulties using screening questionnaires rather than comprehensive clinician-administered psychiatric evaluations. Consequently, the available evidence does not allow us to determine whether elevated PIU scores represented an independent source of impairment, co-occurred with a psychiatric or neurodevelopmental condition, or reflected a coping response to broader psychopathology. Future studies should combine validated and activity-specific measures of problematic digital behavior with developmentally appropriate diagnostic assessments and information obtained from multiple informants.

A further limitation concerns the geographical concentration of the evidence: 13 of the 18 included studies (72.2%) were conducted in China, Türkiye, or Spain. Consequently, the synthesis may disproportionately reflect the sociocultural, educational, and technological contexts of these three countries. Cross-national differences in educational organization, academic expectations and pressure, access to and ownership of digital devices, family rules, social expectations concerning children's autonomy and independent Internet use, family norms and parental mediation practices, peer dynamics, digital infrastructure, platform availability, and regulatory environments may influence both children's patterns of technology use and the ways in which PIU is conceptualized, reported, and measured. Associations observed within particular educational or cultural contexts may therefore not operate in the same way in other regions. The findings should not be interpreted as providing a globally representative account of PIU among primary school children. Further research involving underrepresented geographical regions, together with cross-cultural studies using comparable and developmentally appropriate measures, is needed.

An additional limitation is that none of the studies included in this review examined community-level factors, indicating a gap within the focused body of literature identified for the present narrative synthesis. This finding should not be interpreted as demonstrating that community-level factors have not been examined in the broader literature. Future research should therefore further examine the characteristics of the environments in which children develop beyond their families and schools. The previous literature suggests that neighborhood characteristics, access to safe public and green spaces, and opportunities to participate in recreational and social activities may facilitate in-person play, well-being, and peer interaction, thereby representing potential contextual resources for promoting a more balanced range of leisure activities [84,85]. It is also plausible that access to sporting, cultural, and recreational activities, together with safe community environments that facilitate peer interaction, may help reduce screen exposure and promote healthy leisure alternatives. However, these potential community-level resources were not directly examined in relation to PIU in the studies constituting the central body of the present narrative synthesis. Implementation challenges should also be considered. Children's increasing interest in digital entertainment, together with limited access to suitable spaces and resources, may make efforts to reduce screen time more difficult [82]. Moreover, reductions in screen time should not automatically be interpreted as reductions in PIU, as these constructs are conceptually distinct.

An additional limitation concerns the developmental heterogeneity of the primary school population. The included evidence covered children at different points in the primary school period, but the findings could not be consistently synthesized by age or narrower developmental stage. Younger children and preadolescents may differ substantially

in device ownership, independence of Internet use, parental monitoring, peer influence, cognitive and emotional self-regulation, and their ability to understand and report problematic behavior. Moreover, the reliability and meaning of child self-report measures may vary across this developmental range. Consequently, the associations identified in this review cannot be assumed to operate uniformly among younger and older primary school children, which limits the generalizability of the conclusions across this broad educational stage. Future research should report age- and grade-stratified findings and examine whether the factors associated with higher or lower levels of the specific Internet-related outcomes assessed vary across narrower developmental periods.

In conclusion, the focused body of evidence synthesized in this narrative review highlights the need for further research on PIU using more robust methodological designs, particularly longitudinal studies, as well as more consistent and standardized assessment instruments. These advances would facilitate comparisons across studies, provide a more precise understanding of the processes potentially associated with PIU, and support the design of preventive strategies and interventions tailored to the needs of children.

4.5. Clinical and Practical Implications

Problematic Internet use is a complex phenomenon whose prevention may benefit from considering the different developmental contexts in which children participate. Accordingly, preventive efforts should not be framed solely in terms of replacing or restricting digital activities. Rather, coordinated action by families and schools may support responsible technology use while complementing digital engagement with social, physical, cultural, and recreational activities. However, because most of the evidence synthesized in this review is observational, these possibilities should be regarded as potential directions for the development and evaluation of preventive approaches rather than as interventions of established effectiveness.

Children's increasingly early access to the Internet, video games, smartphones, and other digital devices highlights the potential value of developing and evaluating psychoeducational programs involving students, families, and teachers. Such initiatives could aim to promote healthy and balanced use of digital technologies, strengthen digital literacy and broader digital competencies, provide information about the risks and benefits of online environments, and encourage developmentally appropriate patterns of use. Providing children with opportunities for meaningful offline experiences, including social, physical, cultural, recreational, and traditional play activities, may support a more balanced range of daily activities. Nevertheless, the evidence synthesized in this review does not establish that increasing offline activities or reducing time spent online leads to a reduction in PIU.

Screen or Internet-use duration should not be treated as equivalent to problematic Internet use. Reducing use time does not necessarily indicate improvements in impaired control, persistence despite negative consequences, or functional impairment. Similarly, high engagement does not, by itself, constitute PIU when it is not accompanied by clinically or developmentally relevant difficulties. Preventive programs should therefore evaluate changes in functioning, control over use, negative consequences, and the context and purpose of digital engagement rather than relying exclusively on screen-time reduction as an outcome.

The factors associated with higher or lower levels of Internet use and its problematic manifestations in this review may help generate hypotheses for future preventive studies. Parental supervision and mediation, positive family relationships, offline activities, traditional children's games, and school-based psychoeducation may represent reasonable areas for further evaluation; however, the available evidence is insufficient to establish their effectiveness as interventions for PIU. Figure 4 presents a conceptual socioecological

framework that distinguishes factors associated with lower levels of the specific Internet-related outcomes assessed in the included studies from potential contextual resources proposed on the basis of the broader literature. The latter are presented as hypotheses and priorities for future research rather than as empirically established protective factors or intervention targets. Although these proposed contextual resources may inform the future development of preventive approaches, their relevance as intervention targets requires further empirical evaluation.

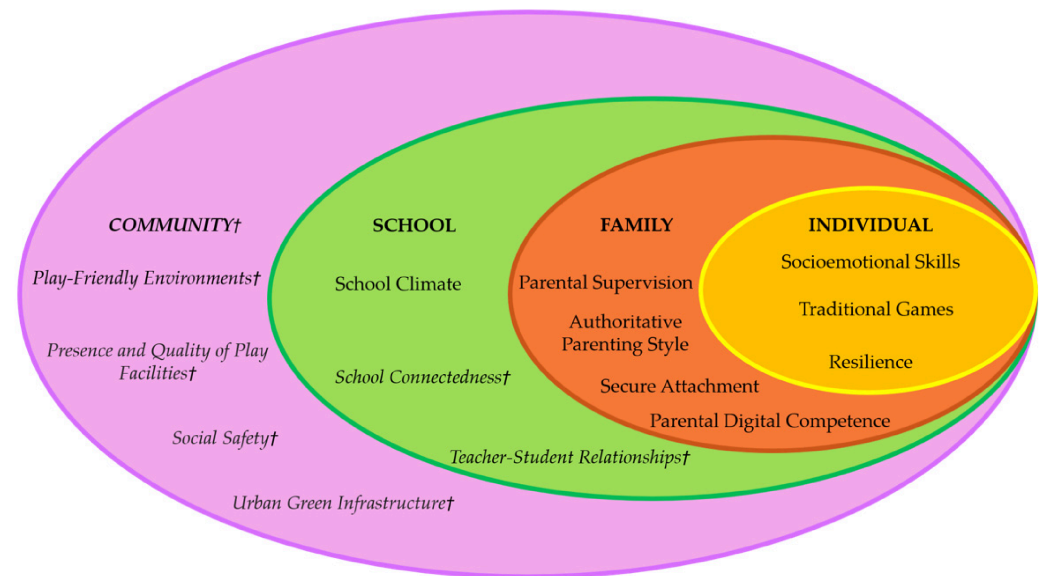


Figure 4. Socioecological Framework of Reported Associations and Proposed Contextual Resources Related to Internet Use and Its Problematic Manifestations among Primary School Children. **Note.** Factors presented in regular type represent associations with lower levels or a lower likelihood of the specific Internet-related outcomes assessed in the studies included in the narrative synthesis. Factors shown in italics and marked with † represent potential contextual resources proposed on the basis of the broader literature discussed in the manuscript; they were not directly examined in relation to these outcomes in the included studies. Only one included study provided school-level evidence, whereas no included study examined community-level factors. Accordingly, the proposed resources should not be interpreted as demonstrating empirically established protective effects or causal relationships.

5. Conclusions

In summary, problematic Internet use (PIU) represents an increasingly relevant area of concern because digital technologies are becoming progressively embedded in the everyday lives of children. The studies constituting the focused body of recent empirical literature examined in this narrative review provide relevant evidence addressing the two objectives established.

Regarding the first objective, the findings reveal considerable terminological and conceptual heterogeneity surrounding PIU in recent empirical research. PIU may serve as an umbrella term encompassing excessive or maladaptive patterns of use across different digital activities, provided that the distinctions among the specific constructs and outcomes are preserved. Furthermore, the constructs identified in the literature can be heuristically organized into broad conceptual categories encompassing adaptive Internet use, risky Internet use, different manifestations of problematic use, and addiction-related constructs, such as Internet addiction and Internet gaming disorder. This organization should not be interpreted as representing sequential stages, a linear progression of severity, or diagnostic equivalence. The categories may occur independently or overlap, and exposure to online

risks, intensity of engagement, impaired control, negative consequences, and functional impairment represent distinct dimensions.

Regarding the second objective, the associations reported across the specific and non-equivalent Internet-related outcomes assessed extended beyond individual characteristics and involved family, socioemotional, peer, school, and digital domains. Positive parent–child relationships, parental supervision, positive peer communication, resilience, participation in traditional children’s games, and a positive school climate were associated with lower levels of or a lower likelihood of the corresponding Internet-related outcomes. Nevertheless, the evidence concerning traditional games included a study assessing general Internet use rather than PIU, while the school-level evidence was derived from only one included study. Conversely, family conflict, authoritarian, permissive, or inconsistent parenting practices, parental technology dependence, peer victimization, insecure parental attachment, and emotional and behavioral difficulties were associated with higher levels of or a greater likelihood of the specific problematic digital-use outcomes assessed. These associations should not be interpreted as demonstrating unidirectional, causal risk-enhancing or protective effects.

In conclusion, this review contributes to clarifying the conceptual diversity surrounding PIU and synthesizes the factors associated with higher or lower levels of Internet use and its problematic manifestations within the focused body of recent empirical literature consulted. However, given the predominance of cross-sectional designs and the conceptual and methodological heterogeneity of the included studies, the findings should be interpreted cautiously and should not be regarded as evidence of causal relationships. The findings support the value of considering a socioecological perspective but primarily identify hypotheses and potential directions for prevention rather than empirically established intervention targets. Evidence from the studies examined is insufficient to determine the effectiveness of specific preventive strategies or interventions. Nevertheless, any future preventive approaches should promote balanced, responsible, and developmentally appropriate digital engagement rather than focusing exclusively on restricting screen time, because the duration of Internet use and PIU are conceptually distinct. Future research should employ longitudinal designs, more standardized and developmentally validated assessment instruments, age- and grade-stratified analyses, and more geographically diverse samples. It should also examine community-level factors, which were not addressed in the focused body of empirical studies synthesized in this review.

Author Contributions: Conceptualization, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; methodology, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; software, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; validation, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; formal analysis, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; investigation, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; resources, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; data curation, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; writing—original draft preparation, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; writing—review and editing, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; visualization, A.N.-V., J.A.C.-R., R.M.L.-G. and C.M.G.-S.; supervision, J.A.C.-R., R.M.L.-G. and C.M.G.-S.; project administration, J.A.C.-R., R.M.L.-G. and C.M.G.-S.; funding acquisition, J.A.C.-R., R.M.L.-G. and C.M.G.-S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data that support the findings of this study are available upon request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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